

Analytical Velocity / Yaw-Rate Limits, Envelope Geometry, and Cross-Axis Coupling of a Mecanum Platform under Steady-State LuGre Friction

Mecanum-PINN dynamics notes

youBot / slip-spin-LuGre simulator (Julia), :lugre_adamov coupling

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Abstract

These notes collect the closed-form results derived for the four-wheel Mecanum platform of the slip-spin-LuGre simulator. We give (i) the 3-DOF equivalent (augmented) plant model the controller and observer act on; (ii) the analytical caps on the longitudinal, lateral, and yaw-rate axes and the physically distinct mechanisms that set them; (iii) the structural shape of the no-slip / trackable velocity envelope in the translational plane ($\dot{\psi} = 0$); and (iv) the origin and quantification of the cross-axis coupling, contrasting the dominant lateral axis, the subdominant longitudinal axis, and the benign yaw axis (treated at a small translational creep v_{creep}). All quantities use the case-1 parameter set. A Word-compatible equation appendix is provided for transcription into the manuscript.

1 System and notation

The platform carries four Mecanum wheels in the O-configuration with roller-axis angles

$$\delta = \left(-\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{4}, -\frac{\pi}{4}\right), \quad \cos \delta_i = \frac{1}{\sqrt{2}} \quad \forall i, \quad \cos^2 \delta_i = \frac{1}{2}. \quad (1)$$

Wheel centres are at $p_x = (h, h, -h, -h)$, $p_y = (l, -l, l, -l)$. The body velocity is $\vec{v} = (V_x, V_y, \dot{\psi})$; the centre of mass is offset from the geometric centre by (a_X, a_Y) . The parameters used throughout are listed in Table 1.

Symbol	Value	Meaning	Symbol	Value	Meaning
h	0.235 m	half-length	μ	0.5	Coulomb coeff.
l	0.15 m	half-width	$s = \mu_s / \mu_c$	1.1	stiction ratio
R	0.05 m	wheel radius	v_s	0.01 m/s	Stribeck vel.
R_a	0.0355 m	roller axle dist.	p_1	0.11	wheel viscous
$R - R_a$	0.0145 m	—	p_2	5.78×10^{-3}	roller viscous
m	30 kg	body mass	$\tau_{\max}$	10 N m	torque cap
m_s	35.6 kg	total mass	a_X	0.016 m	COM offset (long.)
$N_{\text{tot}} = m_s g$	349.2 N	total normal load	a_Y	-0.026 m	COM offset (lat.)
J_w	5.87×10^{-3}	wheel inertia	χ	$\{0, \dots, 0.008\}$	contact-spot

Table 1: Case-1 parameter set (from PlatformParams / LuGreParams).

Contact slip kinematics. With the integrated roller-spin rate $\dot{\gamma}_i$ and the wheel rate ω_i , the per-contact slip velocities are, at the representative roller phase $t_i \rightarrow 0$,

$$V_{px,i} = V_x - \dot{\psi} p_{y,i} - \omega_i R + \dot{\gamma}_i \sin \delta_i (R_a - R), \quad (2)$$

$$V_{py,i} = V_y + \dot{\psi} p_{x,i} + \dot{\gamma}_i \cos \delta_i (R - R_a). \quad (3)$$

The decisive structural fact is that the wheel nulls the body- x slip (its lever is $p_y = \pm l$) and the *passive* roller-spin state $\dot{\gamma}_i$ nulls the body- y slip (its lever is $p_x = \pm h$); the roller inertia $J_{\text{roller}} = 10^{-6}$ is driven solely by contact friction against the viscous drag p_2 .

Wheel and roller dynamics (slip retained, for reference). Away from the no-slip reduction used in §2, the per-wheel and per-roller equations of Adamov & Saipulaev (their Eq. 3, lines 4–5) are kept in full,

$$J_1 \ddot{\varphi}_i = M_i - p_1 \dot{\varphi}_i - F_{x,i} R, \quad (4)$$

$$J_r \ddot{\gamma}_i = -p_2 \dot{\gamma}_i - F_{x,i} \sin \delta_i (R - R_a \cos \varphi_{ei} - \Delta_{Yi} \sin \varphi_{ei} \cos \delta_i) \\ - F_{y,i} \cos \delta_i (R_a - R \cos \varphi_{ei}) - M_{z,i} \sin \varphi_{ei} \cos \delta_i, \quad (5)$$

with the platform rows (x, y, ψ) of Eq. 3 driven by $\sum_i F_{x,i}$, $\sum_i F_{y,i}$, and $\sum_i (F_{y,i} \rho_{Xi} - F_{x,i} \rho_{Yi} + M_{z,i})$, and the contact forces $F_{x,i}$, $F_{y,i}$, $M_{z,i}$ from the LuGre map (6) evaluated at the slip (2)–(3). The motor enters only through (4): it carries $p_1 \dot{\varphi}_i$ and sets $F_{x,i}$, whereas the roller (5) is passive — $\dot{\gamma}_i$ is sustained only by the contact terms against p_2 . This actuated-wheel / passive-roller asymmetry is what splits the caps in §3 and the post-cap behaviour in §6.

Steady-state friction. The LuGre/Stribeck steady force per contact is a bounded vector aligned with the slip,

$$\vec{F}_i = -N_i g_t(\|V_p\|) \frac{V_{p,i}}{\|V_{p,i}\|}, \quad g_t = \mu_c \left[1 + (s - 1) e^{-(\|V_p\|/v_s)^2} \right], \quad (6)$$

so that $\|\vec{F}_i\|$ ranges from $N_i \mu_s$ (static, low slip) down to $N_i \mu_c$ (kinetic). Each contact therefore has an (essentially isotropic) friction circle of radius $\sim N_i \mu$.

Roller-frame force decomposition. Every contact force lives in the plane of its roller, so the natural basis is the roller frame — the axle direction $\hat{e}_i = (\cos \delta_i, \sin \delta_i)$ and the rolling direction $\hat{n}_i = (-\sin \delta_i, \cos \delta_i)$ — in which

$$\vec{F}_i = F_{\parallel,i} \hat{e}_i + F_{\perp,i} \hat{n}_i, \quad F_{\parallel,i}^2 + F_{\perp,i}^2 \leq (\mu N_i)^2. \quad (7)$$

The two components have distinct owners. Along the axle the roller cannot roll, so $F_{\parallel}$ is the *wheel traction*, set by the motor through ω_i (carried with the bearing p_1). Along the rolling direction the roller spins freely against p_2 , so $F_{\perp} = p_2 \dot{\gamma}_i / (R - R_a)$ is the *roller-sustaining drag*, supplied by contact friction because the roller is passive. This is the p_1/p_2 split of §2 made geometric: $F_{\parallel} \leftrightarrow$ motor, $F_{\perp} \leftrightarrow$ contact. Whether a commanded motion requires the roller to spin — hence whether it loads $F_{\perp}$ — is exactly what separates the two translational caps below: a longitudinal command leaves $F_{\perp} = 0$ and the circle empty (a wheel/motor limit), while a lateral command forces $F_{\perp} \neq 0$ and fills it (a roller/contact limit).

2 Equivalent (augmented) plant model

The controller and the disturbance observer are built on a 3-DOF rigid-body *equivalent model* in which the wheel/roller rotational inertias are reflected into the body and the lumped contact dissipation is written as equivalent viscous and Coulomb terms. With $\vec{v} = (V_x, V_y, \dot{\psi})$,

$$M_{\text{aug}} \dot{\vec{v}} + C(\dot{\psi}) \vec{v} + D_{\text{eq}} \vec{v} + \vec{f}_c(\vec{v}) = \vec{F}_p. \quad (8)$$

The augmented mass matrix reflects J_w through the wheel kinematics and keeps the COM coupling,

$$M_{\text{aug}} = \begin{bmatrix} \tilde{m} & 0 & -m a_Y \\ 0 & \tilde{m} & m a_X \\ -m a_Y & m a_X & I_\psi \end{bmatrix}, \quad \tilde{m} = m_s + \frac{4J_w}{R^2}, \quad I_\psi = I_s + \frac{4(l+h)^2}{R^2} J_w. \quad (9)$$

The gyroscopic/centrifugal vector (the only translation–yaw coupling once M_{aug} is formed) is

$$C(\dot{\psi}) \vec{v} = \begin{bmatrix} -m_s \dot{\psi} V_y - m a_X \dot{\psi}^2 \\ m_s \dot{\psi} V_x - m a_Y \dot{\psi}^2 \\ m \dot{\psi} (a_X V_x + a_Y V_y) \end{bmatrix}. \quad (10)$$

D_{eq} is not an independent modelling choice; it is the reflection of the two physical joint frictions into the body, derived next.

Equivalent damping from the no-slip (Model N) reduction

In Adamov & Saipulaev the dissipation lives *only* in the per-wheel/roller equations — base–wheel joint $p_1 \dot{\varphi}_i$ and wheel–roller joint $p_2 \dot{\gamma}_i$ — and the platform equations carry no explicit body damping. To reflect them, impose the no-slip (Model N, non-holonomic) constraints $V_{PiX} = V_{PiY} = 0$ at the central contact $\varphi_{ei} = 0$ and solve for the spin rates in terms of $\vec{v} = (V_x, V_y, \dot{\psi})$:

$$\dot{\gamma}_i = -\frac{V_y + \rho_{Xi} \dot{\psi}}{\cos \delta_i (R - R_a)}, \quad (11)$$

$$\dot{\varphi}_i = \frac{1}{R} \left[V_x - \rho_{Yi} \dot{\psi} + (V_y + \rho_{Xi} \dot{\psi}) \tan \delta_i \right]. \quad (12)$$

With the Rayleigh dissipation $\mathcal{F} = \frac{1}{2} \sum_i (p_1 \dot{\varphi}_i^2 + p_2 \dot{\gamma}_i^2)$, the equivalent body damping is the Hessian $D_{\text{eq}} = \sum_i (p_1 \nabla \dot{\varphi}_i \nabla \dot{\varphi}_i^\top + p_2 \nabla \dot{\gamma}_i \nabla \dot{\gamma}_i^\top)$, where

$$\nabla \dot{\varphi}_i = \frac{1}{R} (1, \tan \delta_i, -\rho_{Yi} + \rho_{Xi} \tan \delta_i), \quad \nabla \dot{\gamma}_i = \frac{-1}{\cos \delta_i (R - R_a)} (0, 1, \rho_{Xi}). \quad (13)$$

Summing over the four wheels ($\tan \delta_i = \mp 1$, $\rho_{Xi} = \pm h$, $\rho_{Yi} = \pm l$) all off-diagonal terms cancel, leaving

$$D_{\text{eq}} = \text{diag} \left(\underbrace{\frac{4p_1}{R^2}}_{d_x}, \underbrace{\frac{4p_1}{R^2} + \frac{8p_2}{(R-R_a)^2}}_{d_y}, \underbrace{\frac{4p_1(l+h)^2}{R^2} + \frac{8p_2 h^2}{(R-R_a)^2}}_{d_\psi} \right) \approx \text{diag}(176, 396, 38). \quad (14)$$

The wheel lever is $(l + h)$, not l : the no-slip constraint folds the roller accommodation into $\dot{\varphi}_i$ (12), so a pure spin turns each wheel at $(l + h)\dot{\psi}/R$ (kinematic gain ≈ 7.7). The same reduction recovers the dissipated power $\sum_i (p_1 \dot{\varphi}_i^2 + p_2 \dot{\gamma}_i^2) = d_\psi \dot{\psi}^2$.

The p_1/p_2 split that drives every cap. Each entry separates into a wheel-joint (p_1) part and a roller-joint (p_2) part, and the two are *supplied by different actuators*. The p_1 parts ($4p_1/R^2$ on x, y ; $4p_1(l + h)^2/R^2 \approx 26$ on ψ) are paid by the *motors*: at steady state $M_i = F_{Xi}R + p_1 \dot{\varphi}_i$, so the bearing drag is covered by motor torque and never loads the friction circle. The p_2 parts ($8p_2/(R - R_a)^2 \approx 220$ on y ; $8p_2 h^2/(R - R_a)^2 \approx 12.2$ on ψ) must be supplied by *contact friction*, since the roller is passive — its spin is sustained only by a contact force bounded by $N_i \mu$. Consequently every *friction-limited* cap (§3) is set by the p_2 part of its axis, while the p_1 part sets the *actuator/power* demand: V_x has no p_2 term (motor-limited, large), whereas V_y and $\dot{\psi}$ are governed by their p_2 parts (friction-limited, small).

The control wrench is the wheel-torque map

$$\vec{F}_p = \left(\frac{1}{R} \sum_i \tau_i, \frac{1}{R} (-\tau_1 + \tau_2 + \tau_3 - \tau_4), \frac{l+h}{R} (-\tau_1 + \tau_2 - \tau_3 + \tau_4) \right), \quad (15)$$

and the Coulomb/LuGre part $\vec{f}_c$ saturates each contact at $N_i \mu$. The caps below are the conditions under which $\vec{f}_c$ or $\vec{F}_p$ saturates on each axis of (8).

3 Velocity and yaw-rate caps

3.1 Longitudinal cap — torque/drag limited ($\dot{\psi} = 0$)

At constant V_x the rollers need not spin ($\dot{\gamma}_i = 0$, hence $F_{\perp,i} = 0$ by (7)), and a wheel rolling at steady speed needs no traction either, so $F_{\parallel,i} \rightarrow 0$: *both* roller-frame components vanish and the friction circle is *empty*. The only steady resistance is the wheel-bearing viscous torque $p_1\omega_i$, carried by the motor, which saturates when it equals $\tau_{\max}$:

$$\boxed{V_{x,\text{cap}} = \frac{\tau_{\max} R}{p_1}} = \frac{10 \times 0.05}{0.11} \approx 4.55 \text{ m/s}. \quad (16)$$

This is a *soft* cap: the circle never binds, so above it the platform merely lags the reference with the contact force still negligible and on-axis. The transient (maneuver) limit is instead set by the aggregate friction circle,

$$a_{x,\text{max}} = \frac{(\sum_i N_i)\mu}{m_s} = \frac{349.2 \times 0.5}{35.6} \approx 4.9 \text{ m/s}^2. \quad (17)$$

3.2 Lateral cap — friction-circle limited ($\dot{\psi} = 0$)

For pure strafe the rollers *must* spin to null $V_{py,i}$, at the rate $\dot{\gamma}_i = -V_y/(\cos \delta_i(R - R_a))$, so each contact must carry the roller-sustaining drag

$$F_{\perp,i} = \frac{p_2 \dot{\gamma}_i}{(R - R_a)} = \frac{\sqrt{2} p_2}{(R - R_a)^2} V_y \quad (\text{along } \hat{n}_i). \quad (18)$$

This $F_{\perp}$ is the *same* on every wheel ($\dot{\gamma}_i$ is fixed by V_y and δ_i , not by N_i), and over the four axles it sums to a pure $-y$ drag $2\sqrt{2} F_{\perp}$. The steady body balance then forces the tractions to supply exactly this in y and nothing net in x , which — by the symmetric $\pm$ pattern over the axles — pins $|F_{\parallel,i}| = F_{\perp,i}$ at every wheel. The two roller-frame components being equal, each wheel's resultant has magnitude

$$\|\vec{F}_i\| = \sqrt{2} F_{\perp,i} = \frac{2 p_2}{(R - R_a)^2} V_y \approx 55.0 V_y \text{ [N]}, \quad (19)$$

directed along $\pm$ body- x (the $\hat{e}_i$ and $\hat{n}_i$ parts cancel in y), alternating in sign over the wheels. It is this body- x resultant — not a body- y force — that loads the circle during strafe. Saturation occurs when it fills the circle, $\sqrt{2} F_{\perp,i} = N_i \mu$. With $N_i = \frac{mg}{4}(1 \pm \frac{ax}{h} \pm \frac{ay}{l}) + m_w g$ and $N_{\min} = N_3 \approx 69.6 \text{ N}$, the lightest wheel saturates first:

$$\boxed{V_{y,\text{crit}} = \frac{N_{\min} \mu (R - R_a)^2}{2 p_2}} \approx 0.63 \text{ m/s}, \quad (20)$$

$$V_{y,\text{agg}} = \frac{(\sum_i N_i) \mu (R - R_a)^2}{8 p_2} \approx 0.79 \text{ m/s}. \quad (21)$$

Since $V_{y,\text{crit}} \propto \mu$, $V_{y,\text{crit}}(\mu) \approx 1.265 \mu = \{0.51, 0.63, 0.76\} \text{ m/s}$ for $\mu = \{0.4, 0.5, 0.6\}$ — the cleanest plant-vs-controller discriminator — and the Stribeck dip places the onset in the hysteretic band $[V_{y,\text{crit}}, s V_{y,\text{crit}}] \approx [0.63, 0.70] \text{ m/s}$.

3.3 Yaw-rate cap: the roller-sustaining drag $F_{\perp}$ fills the circle

A pure spin at $\dot{\psi}$ moves each contact tangentially. Its body- y component $\dot{\psi} \rho_{Xi} = \pm \dot{\psi} h$ must be rolled out by the passive roller, which spins at $\dot{\gamma}_i = -\dot{\psi} \rho_{Xi}/(\cos \delta_i(R - R_a))$, i.e. $|\dot{\gamma}_i| =$

$\sqrt{2} h \dot{\psi} / (R - R_a)$ ((11) at $V_y = 0$).¹ Sustaining that spin against the roller viscous p_2 costs a contact force along the rolling direction $\hat{n}_i$ — the roller-sustaining drag of the roller frame (§1),

$$F_{\perp,i} = \frac{p_2 \dot{\psi}}{(R - R_a)} = \frac{\sqrt{2} p_2 h}{(R - R_a)^2} \dot{\psi} \approx 9.1 \dot{\psi} \text{ N}, \quad (22)$$

the *same* passive force that set $V_{y,\text{crit}}$ (§3.2), now driven by the long spin lever h in place of the strafe unit. It is the only $\dot{\psi}$ -force the contacts *must* supply, and it loads the friction circle directly.

The yaw drag is small; the dissipation is large and wheel-fed. Equation (22) is not in tension with the roller-joint damping

$$d_{\psi}^{(p_2)} = \frac{8 p_2 h^2}{(R - R_a)^2} \approx 12.2, \quad (23)$$

the p_2 part of d_{ψ} in (14). That coefficient is the roller dissipation *power* $d_{\psi}^{(p_2)} \dot{\psi}^2$, whereas the contact *yaw moment* of $F_{\perp}$ is only

$$\sum_i F_{\perp,i} \ell_{\perp,i} = \frac{4 p_2 h (h - l)}{(R - R_a)^2} \dot{\psi} \approx 2.2 \dot{\psi} \text{ N m}, \quad (24)$$

at the short roller-drag lever $\ell_{\perp} = (h - l)/\sqrt{2}$ (§1). The gap between 12.2 and 2.2 is the wheel spin: each roller is turned by *both* the platform spin and the wheel rotation ω_i that nulls V_{px} , and the motors feed the ω_i share. The yaw axis therefore feels only $\approx 2.2 \dot{\psi}$ of contact drag, balanced by an equally small traction mean $\sum_i F_{\parallel,i} \ell_{\parallel} \approx 2.2 \dot{\psi}$ (mean $F_{\parallel} \approx 2 \dot{\psi}$ N); the rest of $d_{\psi}^{(p_2)}$ is delivered through the wheels and never sits on the circle as a yaw drag. $F_{\parallel}$ is the motor/wheel's component — a small driving mean plus a large zero-mean bristle ripple — and $F_{\perp}$ alone sets the cap.

The cap. As $\dot{\psi}$ rises, $F_{\perp}$ climbs (22) and the traction headroom closes,

$$|F_{\parallel,i}| \leq \sqrt{(\mu N_i)^2 - F_{\perp,i}^2}, \quad (25)$$

so the $F_{\parallel}$ band is the full $\pm \mu N_i$ at $\dot{\psi} = 0$ and narrows toward zero as $F_{\perp} \rightarrow \mu N_i$, the lightest wheel ($N_{\min}$, rear-left, diagonally opposite the COM) pinching first. The cap is $F_{\perp} = \mu N_{\min}$:

$$\dot{\psi}_{\max} = \frac{\mu N_{\min} (R - R_a)^2}{\sqrt{2} p_2 h} = \frac{\sqrt{2} V_{y,\text{crit}}}{h} \approx 3.8 \text{ rad/s}. \quad (26)$$

The torque-only rate (≈ 19 rad/s) is never approached: the binding limit is the passive roller force through p_2 , not actuation. This is the simulator signature exactly — $F_{\perp}$ a clean ramp to $\pm \mu N_i$ split by front/rear sign, the $F_{\parallel}$ band collapsing per (25) with the lightest wheel pinched first, the drilling moment negligible.

Why the coupled law caps first. The wall is specific to the coupled LuGre–Adamov law, whose steady force carries the Adamov denominator:

$$\vec{F}_{ss,i} = -N_i \frac{g(s_{t,i})}{s_{t,i}} \vec{V}_{p,i}, \quad s_{t,i} = f_{\text{slip}} \frac{8}{3\pi} |\omega_{z,i}| \chi + \|\vec{V}_{p,i}\|, \quad (27)$$

¹The roller-spin, friction and dissipation relations used here follow Adamov & Saipulaev (2020); see the Reference.

so the spin $\omega_{z,i} = \dot{\psi}$ inflates $s_{t,i}$ and *shrinks* the effective circle radius $\mu_{\text{eff}} N_i$. The coupled law thus reaches $F_{\perp} = \mu_{\text{eff}} N_{\text{min}}$ — the cap (26) — before the *uncoupled* law ($s_{t,i} = \|\vec{V}_{p,i}\|$, no spin dilution), which keeps full μN_i and runs higher: exactly the coupled-walls-first / uncoupled-reaches-3.5 split seen in simulation.

The contact-spot drilling adds only a small χ -dependent Coulomb yaw resistance,

$$M_{\text{spin}} \xrightarrow{v_{\text{creep}} \rightarrow 0} -\mu \left(\sum_i N_i \right) \chi \frac{3\pi}{16} \text{sign}(\dot{\psi}), \quad (28)$$

≈ 0.5 N m at $\chi = 0.005$, negligible beside the roller-sustaining drag.

4 Bounding envelope geometry ($\dot{\psi} = 0$)

The trackable-velocity set in the translational plane is *neither a rhombus nor an ellipse*. Each contact enforces an isotropic friction circle $\|\vec{F}_i\| \leq N_i \mu$; since the slip components (2)–(3) are affine in (V_x, V_y) , each wheel’s limit is an ellipse in (V_x, V_y) and the feasible set is the *intersection of four* oriented ellipses — a rounded convex region, not a single ellipse. A clean rhombus would need scalar one-directional Coulomb caps with a velocity-independent ceiling; the $\|V_p\|$ two-norm and Stribeck exponential in (6) bulge the facets, so no sharp vertices survive. The two axes are also bounded by *different mechanisms* — a near-vertical torque-saturation wall (16) in V_x , a rounded friction fold (20) in V_y — which breaks the 90° symmetry an ellipse would require. A compact description is an anisotropic superellipse

$$\left| \frac{V_x}{V_{x,\text{cap}}} \right|^n + \left| \frac{V_y}{V_{y,\text{crit}}} \right|^n = 1, \quad 1 < n < 2, \quad \frac{V_{x,\text{cap}}}{V_{y,\text{crit}}} \approx 7, \quad (29)$$

with the diamond-like skeleton ($n \rightarrow 1$) from the $\pm 45^\circ$ roller force directions and diagonal de-rating, and the rounding ($n > 1$) from the LuGre smoothing. The exact n requires the numerical fold/eigenvalue tracer.

The envelope is three-dimensional, and friction-model dependent. The (V_x, V_y) region above is the $\dot{\psi} = 0$ cross-section of a full capability body in $(V_x, V_y, \dot{\psi})$. At $\dot{\psi} = 0$ the contact drilling essentially vanishes ($\omega_{z,i} \approx 0$), so the coupled and uncoupled laws agree on this slice; they diverge along the $\dot{\psi}$ axis. The *uncoupled* LuGre caps spin separately from translation, so its body is close to a *product* — the translational region times an independent yaw range — and the cross-section barely contracts as $\dot{\psi}$ rises. The *coupled* LuGre–Adamov law (27) folds the spin $\omega_{z,i} = \dot{\psi}$ into the slip measure $s_{t,i}$, diluting the translational traction, so its body is *pinched*: rising $\dot{\psi}$ eats the translational budget (and rising $\|V\|$ eats the yaw budget), yielding a genuinely coupled convex region that is markedly smaller in the combined-motion corners. The yaw cap (26) and the 40 N body- x wall are this pinch seen on the $\dot{\psi}$ axis. The reachable set in practice is the *coupled* body; the uncoupled one over-predicts capability whenever spin and translation are commanded together.

5 Cross-axis coupling

5.1 Coupling channel

The off-centre COM makes M_{aug} non-diagonal (9), so a force residual on one axis leaks into the others through the off-diagonal entries of M_{aug}^{-1} (nonzero iff $a_X, a_Y \neq 0$). The per-wheel yaw contribution is $M_{z,i} = p_{x,i} F_{y,i} - p_{y,i} F_{x,i} + M_{z,i}^{\text{spin}}$.

5.2 Lateral axis (dominant)

In the roller frame the per-wheel yaw moment splits by component, $M_{z,i} = F_{\parallel,i}\ell_{\parallel,i} + F_{\perp,i}\ell_{\perp,i}$, and for $\cos \delta_i = 1/\sqrt{2}$ each arm collapses to a single magnitude,

$$\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}} \lambda_i \quad (\lambda = (-, +, -, +)), \quad \ell_{\perp,i} = \frac{h-l}{\sqrt{2}} \nu_i \quad (\nu = (+, +, -, -)). \quad (30)$$

The *traction* (parallel) moment therefore acts at the long arm $(h+l)/\sqrt{2} = 0.272$ m and the *roller-drag* (perpendicular) moment at the short arm $(h-l)/\sqrt{2} = 0.060$ m — a $4.5\times$ leverage advantage for $F_{\parallel}$. (The χ -drilling term averages out: $\omega_{z,i}$ oscillates about zero for non-yaw motion, so $\sum_i M_{z,i}^{\text{spin}} \approx 0$.)

The moment is exactly zero up to and including the cap. Below $V_{y,\text{crit}}$ the strafe forces are load-*independent*: $F_{\perp,i}$ is fixed by V_y and δ_i (18) and $|F_{\parallel}| = F_{\perp}$ by balance, so every wheel carries the same magnitude $\sqrt{2} F_{\perp}$ regardless of N_i . The patterns are symmetric and both component moments vanish. At $V_{y,\text{crit}}$ all wheels still sit at the common binding value $\mu N_{\min}$ — the lightest is merely *first* to its limit while the others keep margin — so the set is still symmetric and $M_z = 0$, continuous with the stable regime. There is no step at the cap.

It ramps up only above the cap. Past $V_{y,\text{crit}}$ the lightest wheel is pinned at $\mu N_{\min}$ while the demand keeps climbing on the others, so the forces spread by load and the symmetry breaks; the moment grows from zero. Decomposing the fully saturated state with the front–rear imbalance $\Delta N_{fr} = mg a_X / h \approx 20$ N,

$$\boxed{M_z^{\text{lat}} = \mu mg a_X \frac{l}{h} \approx 1.50 \text{ N m}} = \underbrace{\frac{h+l}{2} \mu \Delta N_{fr}}_{M_z^{\parallel} \approx +1.93} - \underbrace{\frac{h-l}{2} \mu \Delta N_{fr}}_{|M_z^{\perp}| \approx 0.43}, \quad (31)$$

so the weak-lever roller-drag moment *partially cancels* the strong-lever traction moment (the residual is $\propto (h+l) - (h-l) = 2l$, hence the compact $\mu mg a_X l/h$). This is the amplifier: above the cap the passive roller can no longer spin fast enough to sustain $F_{\perp}$, so the *cancelling* term is clipped and the net climbs off 1.50 toward the unopposed traction value $\mu mg a_X (h+l)/(\sqrt{2} h) \approx 2.7$ N m. Thus $M_z^{\text{lat}}(V_y)$ is a *soft ramp* from 0 at $V_{y,\text{crit}}$ to ~ 1.5 – 2.7 N m at deep saturation, routing the front–rear a_X imbalance through $F_{\parallel}$, which the controller must continuously fight and which — via M_{aug}^{-1} — contaminates V_x .

5.3 Longitudinal axis (present but subdominant)

Longitudinal drive needs *no* roller spin ($\dot{\gamma}_i = 0$, $F_{\perp,i} = 0$), so each saturated force lies purely along its axle $\hat{e}_i$ and acts through the full traction arm $(h+l)/\sqrt{2}$ (30) with no roller-drag term to cancel it. With the left–right imbalance $\Delta N_{lr} = mg a_Y / l \approx 51$ N,

$$\boxed{M_z^{\text{lon}} = \mu \frac{h+l}{\sqrt{2}} \Delta N_{lr} = \mu mg a_Y \frac{h+l}{\sqrt{2} l} \approx 6.94 \text{ N m}}, \quad (32)$$

routing the *left–right* a_Y imbalance through $F_{\parallel}$ at the long lever. Its bare magnitude is the largest of the three, yet the longitudinal axis is far less troublesome, for operational reasons: (i) V_x friction-saturates only *transiently*, at $a_x \rightarrow a_{x,\text{max}}$, whereas V_y saturates in steady state at low speed; (ii) the circle is otherwise empty (§3.1), so there is no roller-spin runaway to rotate the force; (iii) it is self-limiting — the kick ends with the acceleration. Hence “not totally absent, but not dominant.”

5.4 Yaw-rate axis: centrifugal creep coupling (benign)

The COM offset makes the yaw rate force the translation — the mirror of the lateral case, and the channel the x, y axes (which carry no observer) must absorb. The centrifugal terms in (10) are a body force

$$\vec{F}_{\text{cf}} = m\dot{\psi}^2 (a_X, a_Y), \quad \|\vec{F}_{\text{cf}}\| = m|a|\dot{\psi}^2, \quad |a| = \sqrt{a_X^2 + a_Y^2} = 0.031 \text{ m}, \quad (33)$$

held by the translational friction until it saturates at

$$\dot{\psi}_{\text{cf}} = \sqrt{\frac{\mu N_{\text{tot}}}{m|a|}} \approx 14 \text{ rad/s}. \quad (34)$$

Because $\dot{\psi}_{\text{cf}} \gg \dot{\psi}_{\text{max}}$ (26), the platform *cannot reach* centrifugal gross slip — the friction cap stops it first. Below the cap the coupling is only a small steady creep balanced by the translational drag,

$$v_{\text{creep}} \approx \frac{m|a|\dot{\psi}^2}{d_t} \approx 1.2 \text{ cm/s at } \dot{\psi} \approx 2, \quad (35)$$

with $d_t = d_x$ or d_y per component. Being *quadratic* in $\dot{\psi}$, it stays about a cm/s near $\dot{\psi} \approx 2$ and grows to only a few cm/s near the cap $\dot{\psi}_{\text{max}}$ — still small, and precisely what the observer-free x, y axes ride out. **Yaw remains the most benign axis: its coupling to (V_x, V_y) is bounded by the unreachable $\dot{\psi}_{\text{cf}}$, unlike the lateral axis whose *linear* load-asymmetry moment is reached at a low speed.**

5.5 Role of the COM offset: imperfect pitchfork

Every coupling moment (31), (32), (33) is $\propto$ the COM offset and vanishes for $a_X = a_Y = 0$; simultaneously M_{aug} becomes diagonal and the leakage channel closes. With a centred COM the strafe-saturation is a *symmetric* state and the relevant bifurcation is a **pitchfork** — the symmetric solution may break to a $\pm$ yaw branch iff the transverse mode (fed by the kinematic $\dot{\psi} e_y$ controller coupling and the gyroscopic $m_s \dot{\psi} V_y$ term) is unstable past $V_{y,\text{crit}}$. The COM offset is the **imperfection parameter** that unfolds this pitchfork into a **saddle-node**, pre-selecting the yaw direction and shifting the threshold. The two falsifiable tests are the μ -sweep (plant vs. controller) and an $a_X, a_Y \rightarrow 0$ sweep (does the unstable transverse eigenvalue survive at zero offset?).

6 Post-cap behaviour: yaw asymmetry and cross-axis leak

6.1 Directional (CW/CCW) asymmetry of the yaw cap

Every friction *drag* in the model is even in $\dot{\psi}$ — the joint terms $p_1 \dot{\phi}_i^2$, $p_2 \dot{\gamma}_i^2$ and the drilling ($\propto \text{sign } \dot{\psi}$) all have $\dot{\psi}$ -independent magnitude — so on their own they cannot split CW from CCW. The COM offset supplies the only odd-in- $\dot{\psi}$ coupling, in two ways.

(i) *Fixed centripetal vs. sign-flipping tangential demand.* Holding the orbiting COM needs a net contact force $\sum_i \vec{F}_i = -m(a_X, a_Y)\dot{\psi}^2$ — even in $\dot{\psi}$ but pinned to the fixed body direction $-\hat{a}$. The yaw-drive friction is tangential, $\hat{t}_i = (-\rho_{Yi}, \rho_{Xi})/r_i$, and *reverses* with spin sense. Each circle must hold

$$\vec{F}_i = \underbrace{\frac{1}{4}m|a|\dot{\psi}^2(-\hat{a})}_{\text{fixed}} + \underbrace{\pm \mu N_i \hat{t}_i}_{\text{flips with } \dot{\psi}}, \quad (36)$$

so the two *add* at a given wheel for one spin sense and partly *cancel* for the other. Since the offset also makes the loads unequal (the lightest wheel N_{min} , diagonally opposite the COM,

binds first), the sense that aligns the centripetal with the tangential demand *at that wheel* saturates it at a lower $\dot{\psi}$ — a lower cap; the opposite sense keeps margin — a higher cap.

(ii) *Gyroscopic coupling.* The yaw row of $C(\dot{\psi})$ (10) contributes $m\dot{\psi}(a_X V_x + a_Y V_y) = m\dot{\psi}|a|(\hat{a}\vec{V})$, linear in $\dot{\psi}$. With the drift $\vec{V}$ forced along $\pm\hat{a}$ by the centrifugal, this is an effective yaw torque that aids one sense and opposes the other — smaller (drift $\sim$ mm/s) but purely antisymmetric, so it adds directly to the split.

The asymmetry scales as the centripetal share of the binding circle, $m|a|\dot{\psi}^2/(4\mu N_{\min}) \approx 10\%$ at $\dot{\psi} \approx 4$ — a CW/CCW difference of a few tenths of a rad/s. It vanishes at $a_X = a_Y = 0$, flips with a mirrored offset, and scales linearly with $|a|$: the same imperfection signature as the lateral cross-coupling.

6.2 Why exceeding $V_{y,\text{crit}}$ bleeds into $\dot{\psi}$ and V_x

The two translational caps fail in *opposite* ways, and that is the whole explanation for why a lateral over-command is dangerous while a longitudinal one is not. $V_{x,\text{cap}}$ is a *soft, torque/drag* limit (§3.1): past it the wheels simply cannot spin faster, *both* roller-frame components stay near zero (§3.1) so the friction circle is empty, and the platform merely *lags* its reference at $V_{x,\text{cap}}$. There is no slip blow-up, no rotating force, nothing to spill — which is exactly the observed behaviour: driving V_x past its cap stays on-axis and quiet.

$V_{y,\text{crit}}$ is a *hard, friction-circle* limit (§3.2): there the body- x resultant $\sqrt{2}F_\perp$ fills the circle. Past it the passive roller can no longer spin fast enough to sustain $F_\perp$, so $V_{py,i}$ is no longer nulled and grows, and the saturated LuGre force both caps in magnitude *and rotates* as the slip turns. Crossing this wall removes the lateral restoring d_y and, through the load spread that opens up above the cap, unbalances the per-wheel forces. That is the trigger, and it feeds both other axes.

Bleed into $\dot{\psi}$. The yaw moment is exactly zero through the cap (symmetric, load-independent forces, §5.2) and *ramps up* only above it: the lightest wheel ($N_{\min}$) pins first, the load spread breaks symmetry, and the roller-spin cap on $F_\perp$ removes the weak-lever term that had been cancelling the strong-lever traction moment. The net M_z^{lat} therefore climbs from 0 at $V_{y,\text{crit}}$ toward $\sim 1.5\text{--}2.7$ N m (§5.2), acting at the long arm $(h+l)/\sqrt{2}$. A platform that was never commanded to spin is thereby forced into $\dot{\psi} \neq 0$ — progressively worse with overshoot, not as a step.

Bleed into V_x . The induced $\dot{\psi}$ re-enters translation by two paths: the Coriolis $-m_s\dot{\psi}V_y$ in the x -row of (10), and the off-diagonal $-ma_Y$ of M_{aug}^{-1} (the COM coupling). With V_y large (past its cap) and $\dot{\psi}$ now nonzero, $m_s\dot{\psi}V_y$ is a substantial body- x force, and V_x drifts off its reference *even though it was never commanded past its own cap*. The contamination runs one way — lateral saturation feeds V_x , never the reverse — because only the lateral cap saturates the circle in the first place. (The *direct* body- x force imbalance vanishes identically — the two diagonals carry equal load, $N_1+N_4 = N_2+N_3$ — so the leak into V_x is precisely this indirect $\dot{\psi}$ path, not a stray x force.)

Why it grows rather than merely couples. With d_y lost, the translation rows keep only the conservative Coriolis,

$$\tilde{m}\dot{V}_x = -d_x V_x + m_s\dot{\psi}V_y, \quad \tilde{m}\dot{V}_y = \underbrace{-d_y V_y}_{\rightarrow 0} - m_s\dot{\psi}V_x, \quad (37)$$

so the skew $m_s\dot{\psi}$ (gain ≈ 142 at $\dot{\psi} = 4$, comparable to the lost $d_y \approx 220$) whirls the growing V_y into V_x and back at frequency $m_s\dot{\psi}/\tilde{m} \approx \dot{\psi}$, dragging V_x along. Two $\dot{\psi}$ -growing forcings turn

the ring into a divergence: the centrifugal $m|a|\dot{\psi}^2$, now unopposed on y , and the gyroscopic feedback $m\dot{\psi}(\hat{a}\cdot\vec{V})$ that returns the leaked translation into the yaw axis — a positive loop. The result is the bounded-then-diverging lateral/yaw oscillation seen once V_y is pushed past $V_{y,\text{crit}}$, with V_x pulled off-axis through the same Coriolis term.

The asymmetry, stated plainly. A longitudinal over-command cannot do any of this: it never saturates the circle, the slip stays small, the force stays aligned with x , d_y is untouched, and no yaw moment is generated — so V_x just rides its soft cap. A lateral over-command does all of it, because $V_{y,\text{crit}}$ is the friction-circle wall, and crossing it at once kills the lateral restoring and — through the COM-driven load imbalance and the rotating saturated force — forces $\dot{\psi}$, which then drags V_x . Falsifiable: the leak onsets at $V_{y,\text{crit}}(\mu)$, scales with the COM offset (a centred COM removes the yaw moment and the centrifugal/gyroscopic legs), and never appears on a pure V_x sweep.

6.3 Exceeding $\dot{\psi}_{\text{max}}$: leak into V_x and V_y

This is the dual of §6.2, reached by over-commanding the yaw axis rather than the lateral one. Past $\dot{\psi}_{\text{max}}$ (26) the lightest wheel (N_{min} , rear-left) can no longer supply $F_{\perp}$: its passive roller cannot spin fast enough to null V_{py} , so that contact slides along $\hat{n}$ (gross slip) and delivers only μN_{min} in place of the demand (22). The front/rear and left/right balance that kept the clean spin translation-free then breaks — $\sum_i F_{\perp,i}\hat{n}_i = 0$ for the symmetric spin, but the capped wheel is short by $\Delta F = F_{\perp}^{\text{dem}} - \mu N_{\text{min}}$, leaving a net body force along the missing contribution,

$$\vec{F}_{\text{leak}} \approx \Delta F \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \quad (\text{forward-right for } \dot{\psi} > 0; \text{ it flips for } \dot{\psi} < 0). \quad (38)$$

As the next-lightest wheel (front-left, also on the light side opposite the COM) binds, its deficit adds a further $+ \text{body-}x$ term while the y -parts of the two left wheels cancel, so the leak rotates toward predominantly *forward* ($+V_x$). Two effects make this a failure rather than a benign drift. First, $F_{\parallel} \rightarrow 0$ at the saturated wheels (25): the small traction mean that drove the spin vanishes — so $\dot{\psi}$ plateaus at $\dot{\psi}_{\text{max}}$ — and the wheels lose the traction authority to oppose the drift, which is uncorrectable at the very wheels producing it. Second, the leaked $\vec{V}$ re-enters through the Coriolis $-m_s\dot{\psi}\vec{V}$ and the COM off-diagonals of (10), feeding a yaw over-command into the translational dynamics it was never meant to touch. The leak onsets at $\dot{\psi}_{\text{max}}(\mu)$, its direction flips with spin sense, and its magnitude scales with the COM offset — a centred COM equalises the loads so all four wheels bind together and (38) vanishes, the same imperfection signature as the lateral channel. It is the likely origin of the observer’s actuator saturation at heading-ramp onset: the ramp drives $\dot{\psi}$ to the cap, the light wheels saturate, a forward disturbance appears, and a single observer bandwidth cannot separate the stiff yaw channel from the soft translational ones — motivating a per-axis split $\omega_{\psi} \gg \omega_{xy}$ and a compensation clip on the saturating channel.

7 Summary of closed-form results

Quantity	Expression	Value (case 1, $\mu = 0.5$)
Longitudinal speed cap	$\tau_{\max} R / p_1$	4.55 m/s
Longitudinal accel. cap	$(\sum N_i) \mu / m_s$	4.9 m/s ²
Lateral onset (1st wheel)	$N_{\min} \mu (R - R_a)^2 / (2p_2)$	0.63 m/s
Lateral full saturation	$(\sum N_i) \mu (R - R_a)^2 / (8p_2)$	0.79 m/s
Yaw-rate cap ($F_{\perp}$ fills circle)	$N_{\min} \mu (R - R_a)^2 / (\sqrt{2} p_2 h)$	≈ 3.8 rad/s
Roller-joint yaw dissipation (p_2)	$8p_2 h^2 / (R - R_a)^2$	12.2 N m s
Contact yaw moment of $F_{\perp}$	$4p_2 h (h-l) / (R - R_a)^2$	≈ 2.2 N m s
CW/CCW cap asymmetry	$m a \dot{\psi}^2 / (4\mu N_{\min})$	$\approx 10\%$
Yaw centrifugal threshold	$\sqrt{\mu N_{\text{tot}} / (m a)}$	14 rad/s
Anisotropy ratio	$V_{x,\text{cap}} / V_{y,\text{crit}}$	≈ 7
Sustained yaw moment (lat.)	ramp $0 \rightarrow \mu m g a_X (h+l) / (\sqrt{2} h)$	$0 \rightarrow 1.5\text{--}2.7$ N m
Transient yaw moment (lon.)	$\mu m g a_Y (h+l) / (\sqrt{2} l)$	6.94 N m
Induced creep (yaw, below cap)	$m a \dot{\psi}^2 / d_t$	≈ 1.2 cm/s ($\dot{\psi}=2$)

A Equation appendix — Word-compatible L^AT_EX commands

Each block below is the linear L^AT_EX string for the corresponding numbered equation, restricted to commands accepted by the Microsoft Word equation editor (LaTeX input mode; for matrices/vectors an alternative UnicodeMath linear form is given). Commands are split across lines only for readability; concatenate them on one line before pasting, or paste line-by-line.

(E1) Roller-axis angles:

$\backslash\delta = (-\backslash\pi/4, \backslash\pi/4, \backslash\pi/4, \backslash-\pi/4), \backslash\text{quad } \backslash\cos^2\backslash\delta_i = 1/2$

(E2) Longitudinal slip ($t_i \rightarrow 0$):

$V_{\text{px},i} = V_x - \backslash\text{dot}\{\psi\} p_{\text{y},i} - \backslash\omega_i R + \backslash\text{dot}\{\gamma\}_i \backslash\sin\backslash\delta_i (R_a - R)$

(E3) Lateral slip:

$V_{\text{py},i} = V_y + \backslash\text{dot}\{\psi\} p_{\text{x},i} + \backslash\text{dot}\{\gamma\}_i \backslash\cos\backslash\delta_i (R - R_a)$

(E4) Stribeck steady friction magnitude:

$g_t = \backslash\mu_c [1 + (s-1) e^{-(|V_p| / v_s)^2}], \backslash\text{quad } s = \backslash\mu_s / \backslash\mu_c$

(E5) LuGre steady force per contact:

$\backslash\text{vec}\{F\}_i = -N_i \backslash, g_t(|V_p|) \backslash, V_{\text{p},i} / |V_{\text{p},i}|$

(E6) Roller-spin force balance:

$p_2 \backslash\text{dot}\{\gamma\}_i = (R - R_a) (F_{\text{y},i} \backslash\cos\backslash\delta_i - F_{\text{x},i} \backslash\sin\backslash\delta_i)$

(E7) Roller spin that nulls lateral slip:

$\backslash\text{dot}\{\gamma\}_i = -V_y / (\backslash\cos\backslash\delta_i (R - R_a))$

(E8) Per-wheel lateral force demand:

$|F_{\text{y},i}| = \backslash\text{frac}\{ 2 p_2 \}{ (R - R_a)^2 } \backslash, V_y$

(E9) Aggregate lateral roller-drag force:

$F_y^{\text{tot}} = \backslash\text{frac}\{ 8 p_2 \}{ (R - R_a)^2 } \backslash, V_y$

(E10) Per-wheel normal loads:

$N_i = \backslash\text{frac}\{\text{mg}\}{4} \backslash\text{left}(1 \backslash\text{pm } \backslash\text{frac}\{a_X\}{h} \backslash\text{pm } \backslash\text{frac}\{a_Y\}{l} \backslash\text{right}) + m_w g$

(E11) Longitudinal (torque/drag) speed cap:

$V_{\text{x},\text{cap}} = \backslash\text{frac}\{ \backslash\tau_{\text{max}} \backslash, R \}{ p_1 }$

(E12) Longitudinal acceleration cap:

$a_{\text{x},\text{max}} = \backslash\text{frac}\{ \backslash\text{left}(\backslash\text{sum}_i N_i \backslash\text{right}) \backslash\mu \}{ m_s }$

(E13) Lateral velocity cap (onset, lightest wheel):

$$V_{y,crit} = \frac{N_{min}}{\mu} \sqrt{(R - R_a)^2}$$

(E14) Lateral velocity cap (full saturation):

$$V_{y,agg} = \frac{\left(\sum_i N_i \right) \mu}{(R - R_a)^2}$$

(E15) Friction-coefficient scaling of the onset:

$$V_{y,crit}(\mu) \approx 1.265 \mu$$

(E16) Stribeck hysteresis band:

$$V_y \in [V_{y,crit}, s, V_{y,crit}]$$

(E17) Axis anisotropy ratio:

$$V_{x,cap} / V_{y,crit} \approx 7$$

(E18) Anisotropic superellipse envelope:

$$\left| \frac{V_x}{V_{x,cap}} \right|^n + \left| \frac{V_y}{V_{y,crit}} \right|^n = 1, \text{quad } 1 < n < 2$$

(E19) Body mass matrix (LaTeX matrix form):

$$M = \begin{bmatrix} m_s & 0 & -m a_Y \\ 0 & m_s & m a_X \\ -m a_Y & m a_X & I_s \end{bmatrix}$$
(E19, Word UnicodeMath linear form -- type then press space):

$$M = [\text{matrix}(m_s \& 0 \& -m a_Y @ 0 \& m_s \& m a_X @ -m a_Y \& m a_X \& I_s)]$$

(E20) Body acceleration:

$$\dot{\vec{v}} = M^{-1} \mathbf{RHS}$$

(E21) Per-wheel yaw contribution:

$$M_{z,i} = p_{x,i} F_{y,i} - p_{y,i} F_{x,i} + M_{z,i}^{spin}$$

(E22) Front-rear normal-load imbalance (wheel weights cancel; body offset only):

$$\Delta N_{fr} = (N_1 + N_2) - (N_3 + N_4) = mg \frac{a_X}{h}$$

(E23) Sustained yaw moment, lateral saturation (roller frame; 0 at cap, ramps above):

$$M_z^{lat} = \mu mg \frac{a_X}{h} \quad (\text{balanced split, } \sim 1.50 \text{ N m})$$

$$M_z^{lat} \rightarrow \mu mg \frac{a_X}{\sqrt{2} h} \quad (F_{perp} \text{ roller-capped, } \sim 2.7 \text{ N m})$$

(E24) Left-right normal-load imbalance (wheel weights cancel; body offset only):

$$\Delta N_{lr} = (N_1 + N_3) - (N_2 + N_4) = mg \frac{a_Y}{l}$$

(E25) Transient yaw moment, longitudinal saturation (force along axle, $F_{perp} = 0$):

$$M_z^{lon} = \mu mg \frac{a_Y}{\sqrt{2}}$$

(E26) Centred-COM limit (instability removed, cap retained):

$$a_X = a_Y = 0 \rightarrow M_z^{lat} = M_z^{lon} = 0$$

(E27) Equivalent (augmented) plant model:

$$M_{aug} \dot{\vec{v}} + C(\dot{\psi}) \vec{v} + D_{eq} \vec{v} + \vec{f}_c(\vec{v}) = \vec{F}_p$$

(E28) Augmented mass matrix (LaTeX matrix form):

$$M_{aug} = \begin{bmatrix} \tilde{m} & 0 & -m a_Y \\ 0 & \tilde{m} & m a_X \\ -m a_Y & m a_X & I_{\psi} \end{bmatrix}$$
(E28, Word UnicodeMath linear form):

$$M_{aug} = [\text{matrix}(\tilde{m} \& 0 \& -m a_Y @ 0 \& \tilde{m} \& m a_X @ -m a_Y \& m a_X \& I_{\psi})]$$

(E29) Reflected inertias:

$$\tilde{m} = m_s + \frac{4 J_w}{R^2}$$

$$I_{\psi} = I_s + \frac{4}{3} (1+h)^2 R^2 J_w$$

(E30) Gyroscopic / centrifugal vector (LaTeX):

$$\begin{aligned} C(\dot{\psi})\vec{v} = & \begin{bmatrix} -m_s \dot{\psi} V_y - m a_X \dot{\psi}^2 \\ m_s \dot{\psi} V_x - m a_Y \dot{\psi}^2 \\ m \dot{\psi} (a_X V_x + a_Y V_y) \end{bmatrix} \\ & \text{(E30, Word UnicodeMath linear form):} \end{aligned}$$

$$\begin{aligned} C(\dot{\psi})\vec{v} = & [\text{matrix}(-m_s \dot{\psi} V_y - m a_X \dot{\psi}^2 \\ & @ m_s \dot{\psi} V_x - m a_Y \dot{\psi}^2 \\ & @ m \dot{\psi} (a_X V_x + a_Y V_y))] \end{aligned}$$

(E31) Equivalent viscous drag (no-slip reduction; p1 motor-supplied, p2 contact-supplied):

$$\begin{aligned} D_{eq} = & \text{diag}\left(\frac{4}{3} p_1 R^2, \right. \\ & \left. \frac{4}{3} p_1 R^2 + \frac{8}{3} p_2 (R-R_a)^2, \right. \\ & \left. \frac{4}{3} p_1 (1+h)^2 R^2 + \frac{8}{3} p_2 h^2 (R-R_a)^2 \right) \end{aligned}$$

(E32) Control wrench (wheel-torque map):

$$\begin{aligned} \vec{F}_p = & \left(\frac{1}{R} \sum_i \tau_i, \right. \\ & \left. \frac{1}{R} (-\tau_1 + \tau_2 + \tau_3 - \tau_4), \right. \\ & \left. \frac{1+h}{R} (-\tau_1 + \tau_2 - \tau_3 + \tau_4) \right) \end{aligned}$$

(E33) Yaw-rate cap (roller-sustaining drag $F_{\perp}$ fills the circle):

$$\begin{aligned} F_{\perp,i} = & \frac{\sqrt{2}}{2} p_2 h \{ (R - R_a)^2 \}, \dot{\psi} \\ \dot{\psi}_{\max} = & \frac{\mu N_{\min}}{\sqrt{2}} (R - R_a)^2 \{ \sqrt{2}, p_2 h \} \\ = & \frac{\sqrt{2}}{2} V_{y,crit} h \\ |F_{\parallel,i}| \leq & \sqrt{(\mu N_i)^2 - F_{\perp,i}^2} \quad (\text{traction headroom}) \\ d_{\psi}(p_2) = & \frac{8 p_2 h^2}{(R - R_a)^2} \quad (\text{roller dissipation power, } \sim 12.2) \\ \sum_i F_{\perp,i}, \ell_{\perp,i} = & \frac{4 p_2 h (h - 1)}{(R - R_a)^2}, \dot{\psi} \\ & (\text{contact yaw moment, } \sim 2.2) \end{aligned}$$

(E34) Contact-spot drilling spin torque (+ creep-free limit):

$$\begin{aligned} M_{\text{spin}} = & - \frac{\mu}{2} (\sum_i N_i) \chi^2 \dot{\psi} \\ & \{ \frac{16}{3} \pi | \dot{\psi} | \chi + 5 v_{\text{creep}} \} \\ M_{\text{spin}} \rightarrow & - \mu (\sum_i N_i) \chi \frac{3\pi}{16}, \\ & \text{sign}(\dot{\psi}) \end{aligned}$$

(E35) Centrifugal body force from COM offset:

$$\begin{aligned} \vec{F}_{cf} = & m \dot{\psi}^2 (a_X, a_Y), \text{quad} \\ | \vec{F}_{cf} | = & m |a| \dot{\psi}^2, \text{quad } |a| = \sqrt{a_X^2 + a_Y^2} \end{aligned}$$

(E36) Yaw centrifugal gross-slip threshold:

$$\dot{\psi}_{cf} = \sqrt{\frac{\mu N_{\text{tot}}}{m |a|}}$$

(E37) Induced translational creep (below cap):

$$v_{\text{creep}} \approx \frac{m |a| \dot{\psi}^2}{d_t}$$

(E38) Centred-COM limit (yaw):

$$a_X = a_Y = 0 \rightarrow \vec{F}_{cf} = 0$$

(E39) Coupled LuGre-Adamov global budget (translation + drilling share one circle):

$$\begin{aligned} | (F_{x,i}, F_{y,i}, M_{z,i}^{\text{spin}}) / \rho | \leq & N_i \mu, g(v_{\text{rel},i}), \\ v_{\text{rel},i} = & \sqrt{v_{x,i}^2 + v_{y,i}^2 + (\rho \omega_{z,i})^2} \end{aligned}$$

(E40) Uncoupled LuGre (separate translational circle + standalone drilling law):

$$\begin{aligned} | (F_{x,i}, F_{y,i}) | \leq & N_i \mu, g(\sqrt{v_{x,i}^2 + v_{y,i}^2}), \\ \text{quad } & M_{z,i}^{\text{spin}} \leq \text{separate drilling law of } \omega_{z,i} \end{aligned}$$

(E41) Emergent yaw damping from joint frictions p1, p2 (via dissipated power):

$$\begin{aligned} \sum_i (p_1 \dot{\varphi}_i^2 + p_2 \dot{\gamma}_i^2) = & \left(\frac{4}{3} p_1 (1+h)^2 R^2 + \frac{8}{3} p_2 h^2 (R - R_a)^2 \right) \dot{\psi}^2 \\ = & d_{\psi}, \dot{\psi}^2 \end{aligned}$$

(E42) No-slip (Model N) spin rates in terms of body velocities:
 $\dot{\gamma}_i = -\frac{V_y + \rho_{X,i}\dot{\psi}}{R - R_a} \cos\delta_i$,
 $\dot{\varphi}_i = \frac{1}{R} \left[V_x - \rho_{Y,i}\dot{\psi} + (V_y + \rho_{X,i}\dot{\psi}) \tan\delta_i \right]$

(E43) CW/CCW yaw-cap asymmetry (fixed centripetal + sign-flipping tangential):
 $\vec{F}_i = \frac{1}{4} m |a| \dot{\psi}^2 (-\hat{a})$
 $\mu N_i, \hat{t}_i, \text{quad } \hat{t}_i = (-\rho_{Y,i}, \rho_{X,i}) / r_i$
 $\text{asymmetry} \sim \frac{m |a| \dot{\psi}^2}{4 \mu N_{\min}}$

(E44) Post-cap translation rotator (friction damping collapsed):
 $\tilde{m}\dot{V}_x = -d_x V_x + m_s \dot{\psi} V_y$,
 $\tilde{m}\dot{V}_y = -d_y V_y - m_s \dot{\psi} V_x$
 $\omega_{\text{leak}} \approx m_s \dot{\psi} / \tilde{m} \approx \dot{\psi}$

(E45) Roller-frame force decomposition (axle e_i , rolling dir n_i ; friction circle):
 $\vec{F}_i = F_{\parallel,i} \hat{e}_i + F_{\perp,i} \hat{n}_i$,
 $\hat{e}_i = (\cos\delta_i, \sin\delta_i)$, $\hat{n}_i = (-\sin\delta_i, \cos\delta_i)$
 $F_{\parallel,i}^2 + F_{\perp,i}^2 \leq (\mu N_i)^2$
 $F_{\perp,i} = p_2 \dot{\gamma}_i / (R - R_a)$ (roller-sustaining, contact-supplied)

(E46) Roller-frame yaw arms (traction lever vs roller-drag lever):
 $\ell_{\parallel,i} = \frac{h+1}{\sqrt{2}}$, λ_i , $\lambda = (-, +, -, +)$
 $\ell_{\perp,i} = \frac{h-1}{\sqrt{2}}$, ν_i , $\nu = (+, +, -, -)$
 $M_{z,i} = F_{\parallel,i} \ell_{\parallel,i} + F_{\perp,i} \ell_{\perp,i}$

(E47) Strafe roller-drag (equal on all wheels) and traction match $\rightarrow$ body-x resultant:
 $F_{\perp,i} = \frac{\sqrt{2}}{p_2} \{(R - R_a)^2\}$, V_y ,
 $|\vec{F}_i| = \sqrt{2}$, $F_{\perp,i} = \frac{2}{p_2} \{(R - R_a)^2\}$, V_y

(E48) Yaw over-command leak ($\psi_{\dot{}} > \psi_{\dot{\max}}$; lightest wheel slides along n):
 $\Delta F = F_{\perp}^{\text{dem}} - \mu N_{\min}$
 $\vec{F}_{\text{leak}} \approx \Delta F, (1/\sqrt{2}, -1/\sqrt{2})$
(forward-right for $\psi_{\dot{}} > 0$; flips for $\psi_{\dot{}} < 0$)

Word notes. (1) In Word, choose *Equation* $\rightarrow$ *Convert* $\rightarrow$ *LaTeX* (or set the input style to LaTeX) before typing. (2) $\dot{\gamma}$, $\vec{F}$, $\tilde{m}$, $\frac{1}{4}$, $\sqrt{2}$, μ , $\approx$, $\sum$, Greek names, and $\hat{\cdot}/_{\cdot}$ all convert directly. (3) For the norm, single bars $| \dots |$ are used for portability; for double bars type $\| \dots \|$ twice. (4) The `bmatrix` forms (E19, E28, E30) work in LaTeX input mode; the `\matrix(...@...)` forms are the equivalent Unicode-Math linear syntax if LaTeX-mode matrices misbehave. (5) Operator names `\mathrm{diag}`, `\mathrm{sign}`, `\mathrm{RHS}` render as upright text; if a build drops them, type the letters in a text run inside the equation.

Reference

Adamov, B. I., & Saipulaev, G. R. (2020). Research on the dynamics of an omnidirectional platform taking into account real design of mecanum wheels (as exemplified by KUKA youBot). *Russian Journal of Nonlinear Dynamics*, 16(2), 291–307.